SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I T Bhupal / 192425243 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:bhupalreddy

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks **/40** ***Signature of the Faculty***

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

1. Subnetting of 192.168.20.0/24

Given Data

- Network address block: 192.168.20.0/24
- Lab A: /26
- Lab B: /27
- Lab C: /28

Step 1: Calculate the Number of Usable Hosts

Formula:

Total IP addresses = 2^(32 − Prefix Length)

Usable host addresses = Total IP addresses − 2

Lab A (/26)

Host bits = 32 − 26 = 6

Total IP addresses = 2^6 = 64

Usable hosts = 64 − 2 = **62**

Subnet mask = 255.255.255.192

Lab B (/27)

Host bits = 32 − 27 = 5

Total IP addresses = 2^5 = 32

Usable hosts = 32 − 2 = **30**

Subnet mask = 255.255.255.224

Lab C (/28)

Host bits = 32 − 28 = 4

Total IP addresses = 2^4 = 16

Usable hosts = 16 − 2 = **14**

Subnet mask = 255.255.255.240

Step 2: Determine the Network Details

Lab A (/26)

- Network Address: 192.168.20.0
- Valid Host IP Range: 192.168.20.1 – 192.168.20.62
- Broadcast Address: 192.168.20.63
- Usable Hosts: 62

Lab B (/27)

- Network Address: 192.168.20.64
- Valid Host IP Range: 192.168.20.65 – 192.168.20.94
- Broadcast Address: 192.168.20.95
- Usable Hosts: 30

Lab C (/28)

- Network Address: 192.168.20.96
- Valid Host IP Range: 192.168.20.97 – 192.168.20.110
- Broadcast Address: 192.168.20.111
- Usable Hosts: 14

Final Answer

Lab	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Lab A (/26)	192.168.20.0	192.168.20.1–62	192.168.20.63	62
Lab B (/27)	192.168.20.64	192.168.20.65–94	192.168.20.95	30
Lab C (/28)	192.168.20.96	192.168.20.97–110	192.168.20.111	14

Conclusion: The three labs are assigned separate, non-overlapping subnets. The remaining unused IP address range is 192.168.20.112 – 192.168.20.255.

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2. Subnetting of Two Systems Using /26

Given Data

System 1 IP Address: 192.168.1.130

System 2 IP Address: 192.168.1.190

Subnet Mask: 255.255.255.192

Prefix Length: /26

Step 1: Calculate the Subnet Size

Host bits = 32 − 26 = 6

Total IP addresses = 2^6 = 64

Usable hosts = 64 − 2 = **62**

Block size = 256 − 192 = **64**

The subnet ranges are:

1. 192.168.1.0 – 192.168.1.63
2. 192.168.1.64 – 192.168.1.127
3. 192.168.1.128 – 192.168.1.191
4. 192.168.1.192 – 192.168.1.255

Step 2: Determine the Subnet of System 1

IP Address: 192.168.1.130

The address falls within 192.168.1.128 – 192.168.1.191.

- Network Address: 192.168.1.128
- Valid Host IP Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191
- Usable Hosts: 62

Therefore, System 1 belongs to subnet **192.168.1.128/26**.

Step 3: Determine the Subnet of System 2

IP Address: 192.168.1.190

The address falls within 192.168.1.128 – 192.168.1.191.

- Network Address: 192.168.1.128
- Valid Host IP Range: 192.168.1.129 – 192.168.1.190
- Broadcast Address: 192.168.1.191
- Usable Hosts: 62

Therefore, System 2 belongs to subnet **192.168.1.128/26**.

Step 4: Verify Subnet Membership

Both IP addresses belong to the same subnet, 192.168.1.128/26.

Hence, both systems can communicate directly at the IP subnet level, provided that no other network restrictions prevent communication.

Final Answer

Details	**System 1**	**System 2**
IP Address	192.168.1.130	192.168.1.190
Subnet Mask	255.255.255.192	255.255.255.192
Network Address	192.168.1.128	192.168.1.128
Valid Host Range	192.168.1.129–190	192.168.1.129–190
Broadcast Address	192.168.1.191	192.168.1.191
Usable Hosts	62	62

Conclusion: Both systems belong to the same subnet, 192.168.1.128/26. Both IP addresses are valid host addresses.

3. Departmental Subnetting of 172.16.0.0/16

Given Data

Allocated Network Block: 172.16.0.0/16

- HR Department: /25
- Sales Department: /26
- IT Department: /27
- Admin Department: /28

The subnets are allocated sequentially using Variable Length Subnet Masking (VLSM), starting from 172.16.0.0.

Step 1: Calculate Usable Host Addresses

HR (/25)

Host bits = 32 − 25 = 7

Total addresses = 2^7 = 128

Usable hosts = 128 − 2 = **126**

Sales (/26)

Host bits = 32 − 26 = 6

Total addresses = 2^6 = 64

Usable hosts = 64 − 2 = **62**

IT (/27)

Host bits = 32 − 27 = 5

Total addresses = 2^5 = 32

Usable hosts = 32 − 2 = **30**

Admin (/28)

Host bits = 32 − 28 = 4

Total addresses = 2^4 = 16

Usable hosts = 16 − 2 = **14**

Step 2: Assign Subnet Addresses

HR Department (/25)

- Network Address: 172.16.0.0
- Subnet Mask: 255.255.255.128
- Valid Host IP Range: 172.16.0.1 – 172.16.0.126
- Broadcast Address: 172.16.0.127
- Usable Hosts: 126

Sales Department (/26)

The next available network address is 172.16.0.128.

- Network Address: 172.16.0.128
- Subnet Mask: 255.255.255.192
- Valid Host IP Range: 172.16.0.129 – 172.16.0.190
- Broadcast Address: 172.16.0.191
- Usable Hosts: 62

IT Department (/27)

The next available network address is 172.16.0.192.

- Network Address: 172.16.0.192
- Subnet Mask: 255.255.255.224
- Valid Host IP Range: 172.16.0.193 – 172.16.0.222
- Broadcast Address: 172.16.0.223
- Usable Hosts: 30

Admin Department (/28)

The next available network address is 172.16.0.224.

- Network Address: 172.16.0.224
- Subnet Mask: 255.255.255.240
- Valid Host IP Range: 172.16.0.225 – 172.16.0.238
- Broadcast Address: 172.16.0.239
- Usable Hosts: 14

Step 3: Determine the Remaining Unused IP Range

The original network block is 172.16.0.0/16.

The last assigned subnet ends at 172.16.0.239.

Therefore, the remaining unused IP address range is:

172.16.0.240 – 172.16.255.255

Final Answer

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR (/25)	172.16.0.0	172.16.0.1–126	172.16.0.127	126
Sales (/26)	172.16.0.128	172.16.0.129–190	172.16.0.191	62
IT (/27)	172.16.0.192	172.16.0.193–222	172.16.0.223	30
Admin (/28)	172.16.0.224	172.16.0.225–238	172.16.0.239	14

Conclusion: The four departments use 240 IP addresses in total, including network and broadcast addresses. The remaining unused range is 172.16.0.240 – 172.16.255.255.

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4. Subnetting of 172.16.10.0/24 Using /27

Given Data

Original Network Block: 172.16.10.0/24

New Subnet Mask: /27

Subnet Mask in Decimal: 255.255.255.224

Given IP Addresses:

- IP Address 1: 172.16.10.78
- IP Address 2: 172.16.10.95

Step 1: Calculate the Total Number of Subnets

Original prefix length = /24

New prefix length = /27

Number of borrowed bits = 27 − 24 = 3

Number of subnets = 2^3 = **8 subnets**

Step 2: Calculate the Usable Hosts per Subnet

Host bits = 32 − 27 = 5

Total addresses per subnet = 2^5 = 32

Usable hosts = 32 − 2 = **30 hosts per subnet**

Block size = 256 − 224 = 32

Step 3: Identify the Subnet of 172.16.10.78

The subnet ranges are:

1. 172.16.10.0 – 172.16.10.31
2. 172.16.10.32 – 172.16.10.63
3. 172.16.10.64 – 172.16.10.95
4. 172.16.10.96 – 172.16.10.127
5. 172.16.10.128 – 172.16.10.159
6. 172.16.10.160 – 172.16.10.191
7. 172.16.10.192 – 172.16.10.223
8. 172.16.10.224 – 172.16.10.255

The IP address 172.16.10.78 belongs to the third subnet.

- Network Address: 172.16.10.64
- Valid Host IP Range: 172.16.10.65 – 172.16.10.94
- Broadcast Address: 172.16.10.95
- Usable Hosts: 30

Step 4: Check the IP Address 172.16.10.95

The address 172.16.10.95 is the broadcast address of subnet 172.16.10.64/27.

Therefore, it falls within the same subnet range but cannot be assigned to an individual host.

Final Answer

- Total Number of Subnets: 8
- Usable Hosts per Subnet: 30
- IP Address: 172.16.10.78
- Network Address: 172.16.10.64
- Valid Host IP Range: 172.16.10.65 – 172.16.10.94

- Broadcast Address: 172.16.10.95

Conclusion: The IP address 172.16.10.95 belongs to the same subnet range as 172.16.10.78, but it is the broadcast address and is not a valid host IP address.

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5. IPv6 Address Compression and Expansion

Given Data

IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix Length: /64

Step 1: Compress the IPv6 Address

IPv6 compression follows two main rules:

1. Leading zeros in each hexadecimal group can be removed.
2. A continuous sequence of zero groups can be replaced by a double colon (::). This notation can be used only once in an address.

Original Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

After removing leading zeros:

2001:db8:0:0:0:ff00:42:8329

After replacing consecutive zero groups with double colon:

Compressed IPv6 Address: 2001:db8::ff00:42:8329

Step 2: Expand the Compressed IPv6 Address

Compressed Address:

2001:db8::ff00:42:8329

The compressed address contains six explicitly written groups. The double colon represents two zero groups, giving a total of eight groups.

After restoring the omitted groups and leading zeros:

Expanded IPv6 Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix

Given prefix length = /64

An IPv6 address contains 128 bits.

Network prefix bits = 64

Interface identifier bits = 128 − 64 = 64

Network Prefix:

2001:db8:0:0::/64

The first 64 bits identify the network, and the remaining 64 bits identify the interface within the subnet.

Step 4: Calculate the Total Number of Possible Interface Identifiers

Formula:

Total possible interface identifiers = 2^(128 − Prefix Length)

= 2^64

= **18,446,744,073,709,551,616**

IPv6 does not use broadcast addresses. This value represents the theoretical number of possible 64-bit interface identifiers, not necessarily the number of ordinary host addresses that can be assigned in practice.

Final Answer

- Original IPv6 Address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Compressed IPv6 Address: 2001:db8::ff00:42:8329
- Expanded IPv6 Address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Prefix Length: /64
- Network Prefix: 2001:db8:0:0::/64
- Interface Identifier Bits: 64
- Total Possible Interface Identifiers: 2^64 = 18,446,744,073,709,551,616

Conclusion: The given IPv6 address compresses to 2001:db8::ff00:42:8329. With a /64 prefix, the first 64 bits represent the network prefix and the remaining 64 bits represent the interface identifier.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation